QUIZ 2(B1)

Name:__________ Roll#____________

Q1:For $g(x) = x - 2\ln(1 + x^2)$ answer each of the following questions.

a) Identify the critical points of the function.
b) Determine the intervals on which the function increases and decreases.
c) Classify the critical points as relative maximums, relative minimums or neither.

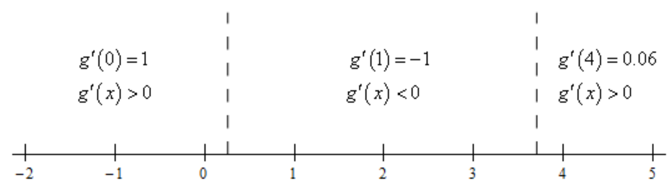
We need the 1st derivative to get the critical points so here it is.

$$g'(x) = 1 - 2\frac{2x}{1+x^2} = \frac{1 - 4x + x^2}{1 + x^2}$$

Now, recall that critical points are where the derivative doesn't exist or is zero. Because we simplified and factored the derivative as much as possible we can clearly see that the derivative will exist everywhere (or at least the denominator will not be zero for any real numbers…). We'll also need the quadratic formula to determine where the numerator, and hence the derivative, is zero. The critical points of this function are then,

$$\boxed{x = 2 \pm \sqrt{3} = 0.2679, \quad 3.7321}$$

To determine the increase/decrease information for the function all we need is a quick number line for the derivative. Here is the number line.



From this we get the following increasing/decreasing information for the function.

$$\boxed{\text{Increasing}: (-\infty, 0.2679) \quad \& \quad (3.7321, \infty) \qquad \text{Decreasing}: \quad (0.2679, 3.7321)}$$

With the increasing/decreasing information from the previous step we can easily classify the critical points using the 1st derivative test. Here is classification of the functions critical points.

$$\boxed{\begin{array}{lcl} x = 0.2679 & : & \text{Relative Maximum} \\ x = 3.7321 & : & \text{Relative Minimum} \end{array}}$$

Q2: Use L'Hospital's Rule to evaluate $\lim_{t \to 0}[\, t . \ln(t + \frac{3}{t})]$

$$\lim_{t \to 0} \left[t \cdot \ln\left(t + \frac{3}{t} \right) \right]$$

$0 \cdot \infty$

$$\lim_{t \to 0} \frac{\ln\left(t + \frac{3}{t} \right)}{\frac{1}{t}}$$

$$\lim_{t \to 0} \left[\frac{\left(\frac{1}{t + \frac{3}{t}} \cdot 1 - \frac{3}{t^2} \right)}{-1/t^2} \right]$$

$$\lim_{t \to 0} \left[\left(\frac{1}{\frac{t^2+3}{t}} \cdot \frac{t^2-3}{t^2} \right) \times -t^2 \right]$$

$$\lim_{t \to 0} \left[\frac{t}{t^2+3} \cdot \frac{t^2-3}{\cancel{t^2}} \cdot -\cancel{t^2} \right]$$

$$\lim_{t \to 0} \left[\frac{t(t^2-3)}{t^2+3} \right]$$

Apply lim

$\boxed{0}$ Ans

QUIZ 2(B2)

Name:__________ Roll#____________

Q1:Use L'Hospital's Rule to evaluate $\lim_{x\to\infty}[e^x + x]^{\frac{1}{x}}$

First, let's define,

$$z = [\mathbf{e}^x + x]^{1/x}$$

and take the log of both sides. We'll also do a little simplification.

$$\ln z = \ln\left([\mathbf{e}^x + x]^{1/x}\right) = \frac{1}{x}\ln[\mathbf{e}^x + x] = \frac{\ln[\mathbf{e}^x + x]}{x}$$

We can now take the limit as $x \to \infty$ of this.

$$\lim_{x\to\infty}[\ln z] = \lim_{x\to\infty}\left[\frac{\ln[\mathbf{e}^x + x]}{x}\right]$$

Before we proceed let's notice that we have the following,

$$\text{as } x \to \infty \qquad \frac{\ln[\mathbf{e}^x + x]}{x} \to = \frac{\infty}{\infty}$$

and we have a limit that we can use L'Hospital's Rule on.

Applying L'Hospital's Rule gives,

$$\lim_{x\to\infty}[\ln z] = \lim_{x\to\infty}\left[\frac{\ln[\mathbf{e}^x + x]}{x}\right] = \lim_{x\to\infty}\frac{\mathbf{e}^x+1/\mathbf{e}^x+x}{1} = \lim_{x\to\infty}\frac{\mathbf{e}^x+1}{\mathbf{e}^x+x}$$

We now have a limit that behaves like,

$$\text{as } x \to \infty \qquad \frac{\mathbf{e}^x+1}{\mathbf{e}^x+x} \to \frac{\infty}{\infty}$$

and so we can use L'Hospital's Rule on this as well. Doing this gives,

$$\lim_{x\to\infty}[\ln z] = \lim_{x\to\infty}\frac{\mathbf{e}^x+1}{\mathbf{e}^x+x} = \lim_{x\to\infty}\frac{\mathbf{e}^x}{\mathbf{e}^x+1} = \lim_{x\to\infty}\frac{\mathbf{e}^x}{\mathbf{e}^x} = \lim_{x\to\infty}(1) = 1$$

Notice that we did have to use L'Hospital's Rule twice here and we also made sure to do some simplification so we could actually take the limit.

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Now all we need to do is recall that,

$$z = \mathbf{e}^{\ln z}$$

This in turn means that we can do the original limit as follows,

$$\lim_{x\to\infty}[\mathbf{e}^x + x]^{1/x} = \lim_{x\to\infty} z = \lim_{x\to\infty}\mathbf{e}^{\ln z} = \mathbf{e}^{\lim_{x\to\infty}[\ln z]} = \boxed{\mathbf{e}}$$

Q2:Use mean value Theorem, Suppose that we know that f(x) is continuous and differentiable on [6,15] Let's also suppose that we know that f(6)=−2 and that we know that $f'(x) \leq 10$. What is the largest possible value for f(15)?

Let's start with the conclusion of the Mean Value Theorem.

$$f(15) - f(6) = f'(c)(15 - 6)$$

Plugging in for the known quantities and rewriting this a little gives,

$$f(15) = f(6) + f'(c)(15 - 6) = -2 + 9f'(c)$$

Now we know that $f'(x) \leq 10$ so in particular we know that $f'(c) \leq 10$. This gives us the following,

$$\begin{aligned} f(15) &= -2 + 9f'(c) \\ &\leq -2 + (9)10 \\ &= 88 \end{aligned}$$

All we did was replace $f'(c)$ with its largest possible value.

This means that the largest possible value for $f(15)$ is 88.